

TicStep RunFile / Record Point Usage

The **Run File** button in the TicStep utility allows you to run a multi-axis path from data defined in a TXT file. The simplest path is just a single multi-axis goal point, where each motor will move from its current position to the goal position, with all axes accelerating, slewing, and decelerating at the same time to achieve a straight-line motion. If multiple goal points are entered, the motors will move from one path point to the next path point in a straight line. If the *segmented flag* is set to 1, the motors will all decelerate to a stop at each path point before accelerating again to the new path point. If the *segmented flag* is set to 0, it is assumed that all of the path points will form a smooth path and the motors will accelerate up to speed at the beginning of the path, move from path point to path point at a constant speed, until finally decelerating to a stop at the final path point.

Before running a multi-axis path, set all motion parameters (current, speed, accel, etc.) for each module involved in the motion. You will also have to set one (and only one) of the modules to be the group leader.

The path TXT file should have the following format:

```
2           : number of axes
3 1         : axis 1 module address, axis 2 module address, ..., axis n module address
100 5       : speed (steps per 1/50 sec.) , acceleration (steps per 1/50 sec per 1/50 sec.)
15 1        : number of path points to execute, segmented flag
4231 17844  : point 1 data – axis 1 coordinate (in steps), axis 2 coord., ....., axis n coord.
8006 33981  : point 2 data – axis 1 coordinate (in steps), axis 2 coord., ....., axis n coord.
.
.
.
```

The file TESTPOINTS_CIRCLE.TXT is an example file that creates a sine/cosine motion (i.e., a circle) with motors 1 and 2.

The **Record Point** button will store positions of the selected modules (in the check boxes to the right) in a single file in a format that can be later run using the **Run File** button. **Record Point** only records a single set of coordinates and automatically sets the *segmented flag* to 1. The speed and acceleration values recorded will be those of the first module selected in the check boxes.

If you want to create more complex motions, you can manually edit a file created with the **Record Point** button, adding additional points from other point data files, and adjusting the number of path points accordingly.